

Q3LOCK: request for independent mathematical review

Review distribution q3lock-v0138-r1 | 8 September 2026

Purpose

We request a critical, location-specific review of Phase Coexistence in an Eight-Component Q3-Locked Quantum Anharmonic Crystal, by Jusang Lee. This is a review edition of research manuscript v0.1.38, not a request for endorsement. The accompanying PDF exists to make external assessment possible. It does not represent a signed external review or a completed journal submission.

The precise object to assess

The manuscript concerns a fixed-spacing, three-dimensional quantum anharmonic lattice with eight real components at each site and a non-radial quartic interaction on the internal cube graph Q3. The source couples to the normalized collective direction. Its proposed conclusion is a conditional composition: under the stated analytic inputs and explicit low-temperature sufficient regime, the infinite-volume pressure has a strict source cusp and there are at least two distinct, parity-related, zero-source tempered Euclidean DLR states.

The author does not assert an all-parameter phase theorem, a necessary phase boundary, pure or extremal states, a complete phase classification, real-time KMS dynamics, a ground-state gap, a continuum limit, a physical vacuum, cosmology, or the closure of a TECT sector. The result remains R-497, T0 and non-claim-bearing in the research registry. No priority claim is made.

What would make your review especially useful

For the mathematics, identify a valid proof, a missing hypothesis, an unjustified limit, or a counterexample at an exact equation or lemma. Please distinguish finite computational identities from infinite-volume and unbounded-operator arguments. For the literature, identify the exact theorem and assumptions if an existing result already subsumes this model and conclusion. A broad title or a similar scalar/radial model is not by itself a subsumption decision.

Partial reviews are welcome. Please identify the rows you actually reviewed and leave every other row NOT REVIEWED. A refusal to certify a step is useful evidence. We do not request a predetermined positive conclusion.

Suggested reviewer profiles

One reviewer with expertise in mathematical statistical mechanics, Euclidean Gibbs/DLR measures, quantum anharmonic crystals and thermodynamic limits can assess the main chain. A complementary reviewer with expertise in unbounded operator forms, infrared bounds and correlation inequalities can examine the collective/Falk-Bruch and reflection/FSS blocks. A specialist familiar with anisotropic vector lattice models should assess novelty and subsumption.

These are expertise criteria, not selected people. No reviewer has been contacted by this packaging operation. Conflicts of interest and availability must be checked before an invitation. An email reply with verifiable attribution and a clear snapshot identification is acceptable project evidence; a specific signature technology is not required.

Efficient reading route

Start with the abstract, model dictionary and conditional main theorem. Then check the manuscript's imported-source comparison and nonclaims. Use the A1-A23 cards in this guide to select a proof block. The archive contains the full theorem-applicability audit, imported-source ledger, literature comparisons, proof-audit document and the original independent-review matrix.

The research snapshot retains older version and PDF-DEFERRED notices for exact replay. The 2026-09-08 review-distribution authorization supersedes the PDF deferral only; it does not supersede mathematical or literature review gates. The PDF's review-edition source is a presentation derivative, not an audited replacement of the original scientific text. The manifest records both hashes.

Reproduction and its limits

Extract the complete archive into a writable directory. First run the included `verify-package.py` script with Python to check every archive member against the manifest. Next enter `research-snapshot` and run the read-only integrated replay command listed in the archive README. Python and SymPy are needed; the tested runtime and dependency versions are recorded in the build report. No source PDF download, Git checkout, TeX engine or network request is needed for that diagnostic replay. TeX is needed only if you choose to rebuild the manuscript.

The replay checks registered finite diagnostics, source identities, assertion coverage and preserved historical output bytes. It is not a formal proof checker for the analytic theorem and does not provide external sign-off.

Return format

Use `review-response.md`. State your name, affiliation if applicable, date, distribution identifier, manuscript hash, expertise and conflicts. For each reviewed A-row give PASS, PASS WITH REPAIR, FAIL or NOT REVIEWED, the precise location, your reason and any affected downstream statements. For repairs, provide replacement text or an explicit missing proposition. State separately whether existing literature subsumes the claim, partially overlaps, or remains unresolved, with exact theorem numbers and hypotheses.

Submission materials are a draft

A journal-neutral cover letter and author checklist are included for later adaptation. No journal has been selected and no submission declarations have been signed. Before submission, the author must decide how review findings affect the theorem, confirm metadata and disclosures, choose the journal, check its current requirements, and approve the final submission. This package does not contact reviewers or transmit material automatically.

Checklist interpretation

The following A1-A23 cards are derived from the original independent-review matrix. Their numbering is not the numbering of TECT Sector A claims. Every card is OPEN for external review at distribution time. Exact manuscript labels are retained so equation numbers can be resolved in the TeX/PDF; a finite diagnostic PASS does not fill in the reviewer's disposition.

A1-A23: load-bearing review cards

A1 | OPEN

Does the eight-component Hamiltonian, source convention, form domain, and fixed-volume self-adjoint realization match the declared model?

Review: Check coercivity, form closure/core, mass convention, and no hidden continuum claim.

Manuscript labels: eq:hamiltonian, eq:form-domain

A2 | OPEN

Does min-max give the absolute heat-trace bound while only the partition function, not a global complex logarithm, is called entire?

Review: Check trace finiteness, complex source majorant, local real analyticity, and the zero-set qualification.

Manuscript labels: eq:trace-bound, eq:source-derivatives

A3 | OPEN

Are the cyclic determinant, Gaussian tightness, residual Riemann-sum limit, and absolute harmonic factor all proved in the stated order?

Review: Check covariance factors, periodic interpolation, Kolmogorov estimate, compact residual passage, and normalized versus absolute laws.

Manuscript labels: eq:cyclic-det, eq:gaussian-weak-limit, eq:weighted-loop-limit

A4 | OPEN

Is the periodic seam count and quartic Young allocation exactly 288, including every endpoint occurrence?

Review: Recount bonds/endpoints and verify the form inequality and trace sandwich.

Manuscript labels: eq:seam, eq:seam-trace-sandwich

A5 | OPEN

Do arbitrary even rectangular exhaustions, subadditivity, periodic transfer, moving-beta secants, and derivative squeezing prove the pressure limit?

Review: Check remainder tiling, finite lower bounds, local uniformity, and no illicit derivative/volume exchange.

Manuscript labels: eq:open-pressure, eq:pressure-seam-rate, eq:pressure-limit

A6 | OPEN

Do KP's general-vector assumptions hold for the non-radial eight-component Q3LOCK potential without importing scalar or radial phase theorems?

Review: Check $\nu=8$, $r_{KP}=2$, finite-range norm, weighted hypotheses, and source line typing.

Manuscript labels: sec:dlr-specification, eq:dlr-potential-envelope

A7 | OPEN

Is source-window exponential moment finiteness established before the Holder recursion is divided?

Review: Check positive normalizers, boundary terms, Gaussian Fernique input, and common constants in h , volume, and state.

Manuscript labels: eq:one-site-exponential, eq:periodic-holder-closure

A8 | OPEN

Is the projective compactness direction correct and is one diagonal compact set obtained for the entire source window?

Review: Check $\alpha_{(k+1)} < \alpha_k$, Holder compact embedding, lower semicontinuity, and cofinality.

Manuscript labels: eq:dlr-compact-set, eq:weighted-tail-direction

A9 | OPEN

Does the finite specification pass to a source-zero DLR limit on a determining class?

Review: Check compact-boundary coercivity, normalizer lower bound, Feller topology, Borel extension, and source-limit order.

Manuscript labels: eq:dlr-normalizer-lower, eq:kernel-source-lipschitz, sec:dlr-source-tangents

A10 | OPEN

Is the log-density mixed Hessian nonnegative with the correct sign for temporal, spatial, and Q3 locking bonds?

Review: Recompute the Q3 identity and verify finite-grid supermodularity without radiality.

Manuscript labels: eq:log-supermodular

A11 | OPEN

Does association pass from finite grids to continuous loops, bounded Borel tests, selected spatial limits, and clipped products?

Review: Check order-preserving wrap interpolation, closed-cone approximation, uniform integrability, and the explicit mixture nonclaim.

Manuscript labels: sec:fkg-loop-passage, eq:fkg-product-clipping

A12 | OPEN

Do Hilbert-valued reflection positivity and the finite-dimensional FSS theorem use exactly the stated finite-mesh prior and edge source?

Review: Check finite-rank Gaussian kernel passage, spatial-only scope, $J=c$, source pairing, and no mesh-uniform prior constant.

Manuscript labels: eq:hilbert-kernel-positive, eq:spatial-reflection-positive, eq:fss-poisson-energy

A13 | OPEN

Does the FSS-to-loop UI passage yield the correctly normalized nonzero-mode Duhamel bound and convergent three-dimensional sum while excluding the zero mode?

Review: Check truncation/UI, beta factors, eigenvalue $2E(p)$, shell tails, and the missing zero-mode lower-bound boundary.

Manuscript labels: eq:fss-source-ui, eq:duhamel-poisson, eq:infrared-bound, eq:infrared-subtraction

A14 | OPEN

Is scalar Jensen valid on the translated form domain and does it give the collective Hessian expectation without differentiating an unbounded trace?

Review: Check translated form-domain invariance, polynomial integrability, spectral Jensen, and the global minimum argument.

Manuscript labels: eq:collective-jensen-chain, eq:collective-hessian

A15 | OPEN

Are the collective form identity, absolute energy-weighted sums, finite spectral cutoffs, and ordered coordinate cutoffs valid?

Review: Check common form core, row/column interchange, nonnegative regrouping, and fixed-R then R-limit order.

Manuscript labels: eq:form-identity, eq:collective-absolute-energy-sums, eq:collective-spectral-cutoff

A16 | OPEN

Does FKG supply exactly the expectation inequalities used in the collective moment chain, rather than pointwise graph inequalities?

Review: Check parity, selected-state scope, coordinate clipping, graph degree, and the direction of every expectation inequality.

Manuscript labels: eq:moment-chain, eq:collective-graph-expectation

A17 | OPEN

Does the squared-tail lemma bound the thermodynamic second moment at a nondifferentiable even pressure?

Review: Check Chernoff signs, fixed-source order, layer-cake integral, and limsup versus liminf use.

Manuscript labels: lem:griffiths-endpoint, eq:griffiths-squared-tail, eq:composition-liminf-limsup

A18 | OPEN

Does the source-window limit preserve the local unbounded expectation in a zero-source DLR state?

Review: Check compact split, kernel continuity, clipped convergence, and no weak-limit passage for an unbounded observable without UI.

Manuscript labels: eq:dlr-local-clipping, eq:source-zero-tangent, eq:mu-plus

A19 | OPEN

Are the conclusions restricted to the strict sufficient regime and do they avoid identifying an exact critical temperature?

Review: Check $r < 0$, $A_0 > I_3$, $\beta > \beta_*$, nonempty parameter set, equality case, and undecided complement.

Manuscript labels: eq:threshold-algebra, eq:beta-star, eq:cusp

A20 | OPEN

Are citations, nonclaims, and the two branch limit orders synchronized throughout the manuscript and literature comparison?

Review: Check no scalar/radial shortcut, no priority claim, no KMS/ground-state/continuum overclaim, and exact authority chain.

Manuscript labels: sec:composition-dictionary, sec:crosswalk

A21 | OPEN

Are the half measures finite and is the Hilbert-kernel proof valid for bounded Borel spatial reflection tests?

Review: Check residual allocation, crossing-bond factorization, finite complex pushforward, and bounded-Borel extension.

Manuscript labels: eq:rp-local-measure, eq:hilbert-kernel-positive, eq:spatial-reflection-positive

A22 | OPEN

Does the finite logarithmic-mean proof give the exact Falk-Bruch normalization, including degenerate energies and $c=0$?

Review: Check concavity signs, weights, energy degeneracy, homogeneity, and endpoint continuity.

Manuscript labels: eq:falk-phi-concavity, eq:falk-spectral-weights, eq:finite-falk-bruch

A23 | OPEN

Does parity preserve the full zero-source specification and does a bounded local observable distinguish the two constructed states?

Review: Check pushforward direction, tempered-space invariance, local clipping bound, and strict witness inequality.

Manuscript labels: eq:parity-specification-intertwining, eq:bounded-phase-witness

Decision sheet and next steps

Identify the snapshot

Give distribution ID, manuscript PDF hash, original source hash, date and reviewer attribution. The editable review-response.md contains all fields.

Keep dispositions separate

Report mathematics and literature separately. Record PASS, PASS WITH REPAIR, FAIL or NOT REVIEWED for every reviewed row, with equation numbers and downstream consequences.

No default acceptance

Every blank or unreviewed row stays open. A compilation, manifest check or successful finite replay is not an analytic or literature review.

Return actionable criticism

Provide an explicit replacement statement, missing estimate or exact subsuming theorem. State uncertainty and coverage limits. Negative or partial findings are welcome.

Submission remains author-controlled

The cover letter is an unsent template. Author declarations, target journal, current journal requirements and final approval remain unresolved. No transmission occurs as part of this package.